

Question Number	Answer	Additional Guidance	Mark
1(a)(i)	An answer that includes two of the following points • {position / location} of (cells / tissues) within (bundles / stem) (1) • size (of cells) / cell wall thickness (1)	Ignore incorrect names Accept pattern	(2)

Question Number	Answer	Additional Guidance	Mark
1(a)(ii)	An answer that includes the following points: • B (1) • phloem (1) • the cells {are sieve tubes / have (sieve) plates / end walls} (1)	Accept partitioned	(3)

Question Number	Answer	Additional Guidance	Mark
1(a)(iii)	An answer that includes the following steps: - measurement of cell {diameter / radius} in epg units and conversion to m (1) - calculation of area using πr^2 (1)	ACCEPT 14 to 14.5 as alternatives to 13. e.g. $\{13 / 6.5\} \times 3 \times 10^{-6} = 39 \times 10^{-6} = 3.9 \times 10^{-5}$ OR $19.5 \times 10^{-6} = 1.95 \times 10^{-5}$ $\pi \times ((1.95 \times 10^{-5})^2) = 1.195 \times 10^{-9}$ OR $3.14 \times 3.8025 \times 10^{-10} = \underline{1.194 \times 10^{-9}}$ ACCEPT <u>1.195, 1.19, 1.2</u> If include units must match answer, but units not needed	(2)
Question Number	Answer	Additional Guidance	Mark
1(a)(iv)	division of area of F by area given of E (1)	ecf from 1a(iii) eg $1.194 \div 0.0613 = 19.5$ times	(1)

Question Number	Answer	Additional Guidance	Mark
1(a)(v)	An description that includes the following points: • description of cutting thin section (1) • use of {stain / dye} (1) • place section (on slide) under coverslip (1) • draw under low power / described (1)	e.g use of razor, botanical razor, sharp knife, scalpel, cutting transverse, horizontal ignore name NOT HP for drawing, ignore if they go on to HP so long as clear they would draw under LP.	(3)

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	A drawing showing the following features: • correct cell outline and ratio width to length (1) • correct outline, dimension and location of nucleus with nucleolus (1) • three correct labels (1)	Ignore cytoplasmic inclusions e.g., longer than wide e.g., top and about half total area nuclear pores drawn negate this mark if they <i>just</i> draw nucleus judge proportions and nucleolus position for mp2 e.g. correct outline, dimension and location of nucleolus	(3)

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	An answer that includes the following points: Difference: - nuclear envelope double membrane, cell membrane single OR {nuclear pore present in nuclear envelope, not in cell membrane / cell membrane has channels} (1) Similarity: - both membranes are phospholipid (bilayer) (1)	Accept functional aspects, e.g. Difference: description of transport of different substances Similarity: both control movement of {substances / relevant named} NOT lipoprotein	(2)